

# A language for websites — and the paper they print on.

A file describes what a page shows: its elements, the state behind them, what happens when the reader acts. The compiler turns it into a static site, an app, a PDF or a deck.

```
terminal

$ wf init hello --template spa
Created hello/ with AGENTS.md
$ wf build
Building hello...
app.js 4.6 kB gzipped
0 errors, 0 warnings
$ wf build -d deck
Slides: 16 slide(s)
```

static site

SPA

PDF

slides

custom elements

## THE IDEA

# One file is the page.

### WHAT YOU WRITE

Home.wf

```
src/pages/Home.wf

page Home(path: "/", title: "Home") {
  state count = 0
  derived doubled = count * 2

  Container {
    Heading("Welcome").h1
    Text("Count: {count} · doubled {doubled}")
    Button("+1").primary {
      on click { count = count + 1 }
    }
  }
}
```

### WHAT THE READER GETS

index.html · app.js



localhost:3000/

## Welcome

Count: 3 · doubled 6

3 +1

The text repaints when the state changes — no virtual DOM, no hooks, no build config.

01

# The language

Declarations are lowercase, elements capitalised, and anything that resolves to nothing is an error — never a silent no-op.



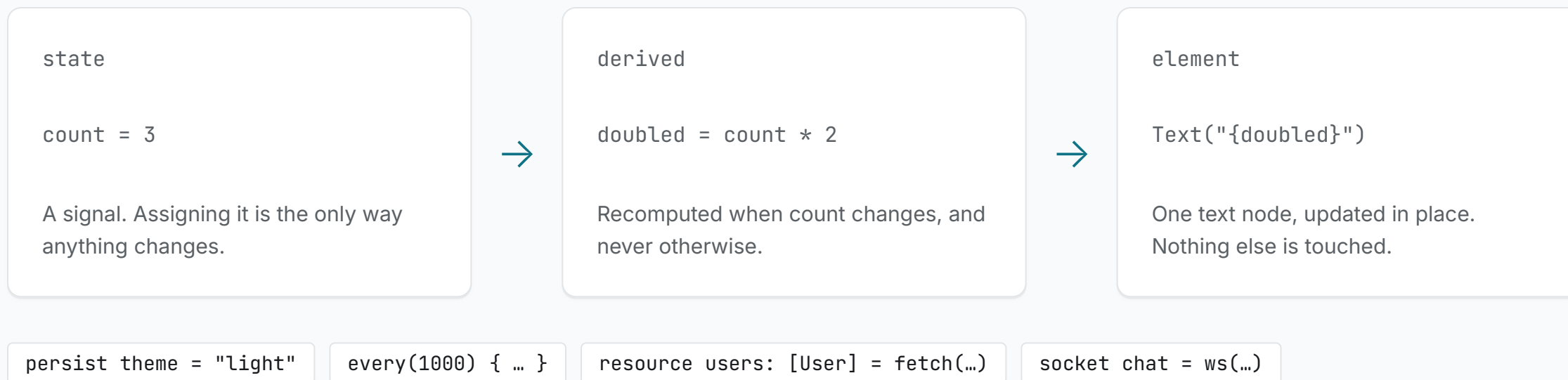
# One lexeme, one meaning.

Seven shapes cover everything a page says. The compiler, the linters and the editor read one registry of what each element takes.

| WRITTEN                               | WHAT IT IS                                                          |
|---------------------------------------|---------------------------------------------------------------------|
| <code>page · component · store</code> | <b>A declaration</b> — a lowercase keyword names what is being made |
| <code>Button · Card · Table</code>    | <b>An element</b> — capitalised, drawn from one registry            |
| <code>.primary .lg .h1</code>         | <b>A flag</b> — a boolean prop, or one case of an enum prop         |
| <code>label: "Name"</code>            | <b>A prop</b> — passed by name; one positional may lead             |
| <code>Card { ... }</code>             | <b>A block</b> — the element's children                             |
| <code>on click { ... }</code>         | <b>A handler</b> — what happens when the reader acts                |
| <code>style { padding: \$md }</code>  | <b>Raw CSS</b> — with design tokens and live {values}               |

# State is the only thing that moves.

What reads a state repaints when it changes. A derived value follows the values it reads; an effect runs when they do.



## TYPES

# The types a page needs, built in.

Each is a plain JSON value at run time, so it crosses a fetch, storage and the static paint unchanged.

### Date

`@2026-03-14`

`.plus(days: 3)`

### Time

`@09:30`

`.plus(minutes: 30)`

### DateTime

`@2026-03-14T09:30Z`

`.inZone("Europe/Berlin")`

### Duration

`90.minutes`

`.hours()`

### Money

`€12.99`

`.convert(rate, "USD")`

### Url

`"https://..."`

`.with(query: ...)`

### Email

`"ada@example.com"`

`.domain()`

### Color

`#0F766E`

`.mix(other, 0.2)`

### Uuid

`uuid()`

a string's methods

### File

`FileUpload`

`.preview()`

### Number(1..=30)

`state nights = 12`

a condition, checked

### Secret

`T12 if it escapes`

never shown, logged or kept

# A mistake is found where it is written.

```
error[T05]: `User` has no field `nmae`  
  → src/pages/Profile.wf:3:18  
  |  
3 |   Heading(user.nmae).h1  
  |               ^^^^  
  = help: Its fields are `id`, `name`  
  = fix:  nmae → name
```

**116 codes, one registry**

The build, the editor, wf explain and the guide read the same list.

**fix A fix where one is known**

The nearest name, a ?., the missing props — applied by the editor.

**1× Every mistake, once**

Every file is read even when another does not parse.

02

# What it builds

One program; five outputs. The same elements are a page in a browser and a page on paper.

---

# Pick the output in one line of config.



## Static site

Every page pre-rendered, then hydrated. Sitemap, canonical, Open Graph.

```
"ssg": true
```



## Single-page app

A page per chunk, loaded when its route shows; a service worker on request.

```
"spa"
```



## PDF

The page's own CSS, laid out on paper: running headers, contents, charts.

```
"pdf"
```



## Slide deck

A slide to a page, from components and data — like this one.

```
"slides"
```



## Custom elements

Components as tags for React, Vue, Rails or a plain page.

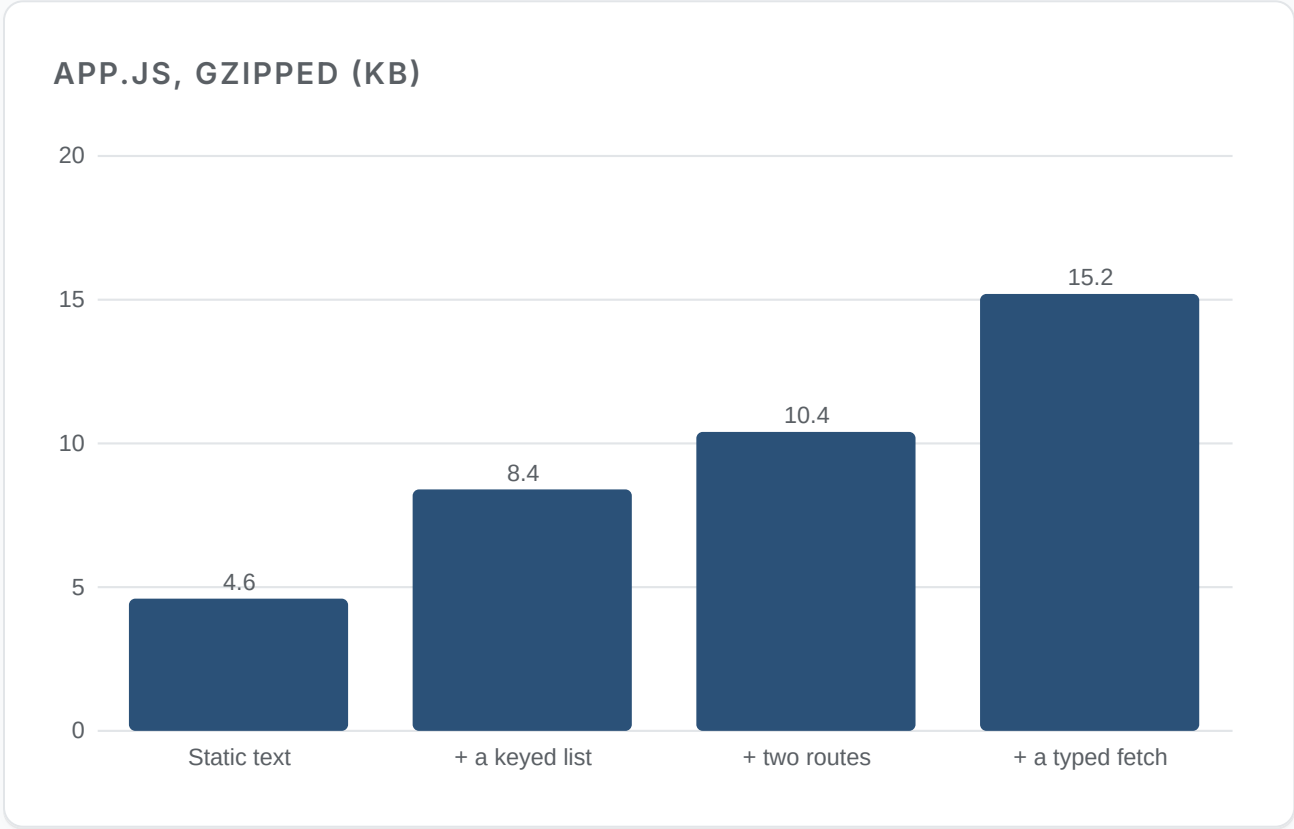
```
"elements"
```

webfluent.app.json

```
{ "build": { "output_type": "slides",  
             "slides": { "size": "16:9" } } }
```

The program does not change. A PDF is refused what only a browser can do — a button, a handler, a route — where it is written, and the editor offers only what the output draws.

# The runtime holds only what a program reaches.

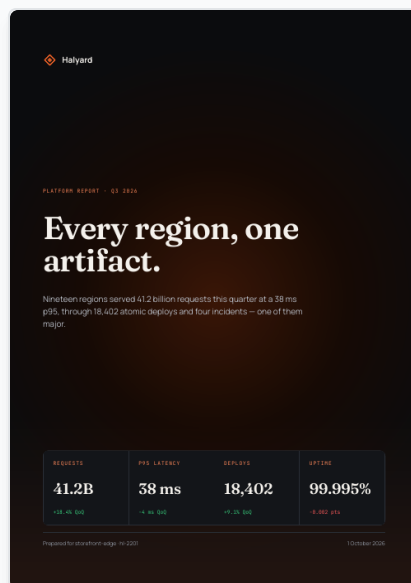


|                 |         |
|-----------------|---------|
| Static text     | 4.6 kB  |
| + a keyed list  | 8.4 kB  |
| + two routes    | 10.4 kB |
| + a typed fetch | 15.2 kB |

Measured on four programs built with wf build. A for brings the keyed-list module, a second page the router, a resource the network engine.

# The same CSS, on paper.

No browser, no external tool: the compiler lays out the page's HTML and CSS itself — the cascade, flex and grid, shaped Arabic, tables that repeat their header.



Report cover — a radial glow and stat tiles



Charts side by side, a table by content

Report: Performance report 20 2020

Workflow: Page 1 of 3

## Deployments

The party four most recent deploy to production. A deploy is one content-addressed build reaching every region at once, or once in a deploy of an older hash.

| Entity  | Content                         | Source        | Artifacts | Time | Status              | Notes            |
|---------|---------------------------------|---------------|-----------|------|---------------------|------------------|
| 1681888 | Move image version to workers   | edge-cache    | barbosa   | 64s  | <span>Failed</span> | 2020-09-28 12:27 |
| 1681834 | Move image version to workers   | edge-cache    | margaret  | 77s  | <span>Failed</span> | 2020-09-28 13:18 |
| 1681829 | Move localized build to worker  | edge-cache-v2 | ada       | 99s  | <span>Failed</span> | 2020-09-28 17:35 |
| 1724474 | Split checked build             | ie-edgecache  | barbosa   | 276s | <span>Failed</span> | 2020-09-29 20:46 |
| 7032161 | Route DC:PRM to fail            | edge-cache    | ada       | 188s | <span>Failed</span> | 2020-09-29 14:34 |
| 7032125 | Route DC:PRM to fail            | edge-cache    | margaret  | 109s | <span>Failed</span> | 2020-09-29 18:53 |
| 4477704 | Split alert on-grid             | edge-cache    | barbosa   | 144s | <span>Failed</span> | 2020-09-28 14:29 |
| 1321437 | Move image version to worker    | ie-edgecache  | brat      | 144s | <span>Failed</span> | 2020-09-28 18:17 |
| 4750428 | Route image version to worker   | checked-v2    | barbosa   | 142s | <span>Failed</span> | 2020-09-28 15:13 |
| 4750434 | Route TUI options for PCJ       | edge-cache    | barbosa   | 180s | <span>Failed</span> | 2020-09-27 18:45 |
| 7494121 | Upgrade cache keys on /api/call | edge-cache    | barbosa   | 244s | <span>Failed</span> | 2020-09-27 04:28 |
| 1725275 | Split checked build             | checked-v2    | ada       | 58s  | <span>Failed</span> | 2020-09-27 14:42 |
| 4864764 | Split alert on-grid             | edge-cache    | barbosa   | 118s | <span>Failed</span> | 2020-09-26 20:39 |
| 7093248 | Send rates to the fail          | edge-cache    | margaret  | 132s | <span>Failed</span> | 2020-09-26 12:18 |
| 4987354 | Route DC:PRM to fail            | ie-edgecache  | brat      | 187s | <span>Failed</span> | 2020-09-24 12:38 |
| 4464705 | Upgrade PRM at the edge         | edge-cache    | ada       | 38s  | <span>Failed</span> | 2020-09-25 13:49 |
| 4864703 | Run updates to DSA              | edge-cache    | ada       | 121s | <span>Failed</span> | 2020-09-25 08:39 |
| 4645202 | Route image version to worker   | ie-edgecache  | margaret  | 64s  | <span>Failed</span> | 2020-09-25 12:44 |
| 8715700 | Send rates to the fail          | ie-edgecache  | margaret  | 180s | <span>Failed</span> | 2020-09-24 16:58 |
| 4717700 | Route image version to worker   | checked-v2    | ada       | 87s  | <span>Failed</span> | 2020-09-24 04:23 |
| 4788148 | Upgrade localized build         | checked-v2    | brat      | 47s  | <span>Failed</span> | 2020-09-24 09:58 |

Fetch data from the report base, not from product.

Page 2 of 3

64 rows, the header on every page



Meridian Studio

Product design and engineering

Invoice

No. 2024-0417

FROM

Meridian Studio  
180 Hudson Ave, Suite 4  
Portland, OR 97209  
ENR-6-02557

BILL TO

Northwest Outfitters, Inc.  
44N. Accounts Payable  
88 Marston Drive, 6th floor  
New York, NY 10012

Issued

Terms

Due

Sep 30, 2026

New 30

96-55332

Oct 30, 2026

| #  | DESCRIPTION                                           | QTY | UNIT         | PRICE      | AMOUNT     |
|----|-------------------------------------------------------|-----|--------------|------------|------------|
| 1  | Component library (Prpm)<br>Change request QR-1       | 2   | component    | \$160.00   | \$320.00   |
| 2  | Illustration set<br>Sept 1-5                          | 1   | illustration | \$330.00   | \$330.00   |
| 3  | Performance budget review<br>Sept 1-5                 | 1   | day          | \$1,100.00 | \$1,100.00 |
| 4  | Copy editing<br>Change request QR-1                   | 1   | hour         | \$65.00    | \$65.00    |
| 5  | Wireframes, checkout flow<br>Sept 1-5                 | 1   | screen       | \$220.00   | \$220.00   |
| 6  | Front-end build, product pages<br>Sept 1-5            | 8   | day          | \$1,180.00 | \$9,440.00 |
| 7  | Wireframes, checkout flow<br>Change request QR-1      | 1   | screen       | \$220.00   | \$220.00   |
| 8  | Front-end build, product pages<br>Change request QR-1 | 3   | day          | \$1,180.00 | \$3,540.00 |
| 9  | UX research interviews<br>Phase 2                     | 3   | session      | \$580.00   | \$1,740.00 |
| 10 | Illustration set<br>Change request QR-1               | 1   | illustration | \$330.00   | \$330.00   |
| 11 | Copy editing<br>Sept 1-5                              | 8   | hour         | \$85.00    | \$680.00   |
| 12 | Wireframes, checkout flow<br>Change request QR-1      | 1   | screen       | \$220.00   | \$220.00   |

Meridian Studio, 180 Hudson Ave, Portland, OR 97209 | 503-639-0000 | [hello@meridianstudio.com](mailto:hello@meridianstudio.com)

Invoice 2024-0417 | page 1 of 1

An invoice: band, watermark, totals kept whole



Arabic, right to left, from the same engine

03

# The tools

One binary: a compiler, a formatter, a test runner, a browser check, an audit — and a language server that knows what you are building.

---

# Everything a project needs is a subcommand.

|                  |                                                                                  |                 |                                                                             |
|------------------|----------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------|
| <b>wf check</b>  | Every finding, with a code, a place and a fix; nothing written                   | <b>wf test</b>  | test <code>"..." { }</code> blocks — snapshots, and clicks in a real Chrome |
| <b>wf verify</b> | Every route loaded in a headless browser: errors, paint, bytes                   | <b>wf audit</b> | What the project trusts: markup, origins, storage, env                      |
| <b>wf fmt</b>    | One layout for every file — <code>.wf</code> braces or <code>.wfx</code> indents | <b>wf docs</b>  | A gallery of every built-in and the project's own components                |
| <b>wf serve</b>  | Rebuild on save; a failed build drawn over the page                              | <b>wf-lsp</b>   | Completion, hover, rename, quick fixes — for this output                    |

# In numbers.

|                                                                |                                                               |                                                         |                                                                      |
|----------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------|
| <b>79</b><br>built-in elements, from Button to TableOfContents | <b>116</b><br>diagnostic codes, each with a page in the guide | <b>37</b><br>runtime modules, shipped only when reached | <b>48</b><br>chapters of the guide, every example built by the tests |
| <b>5</b><br>outputs from one program                           | <b>11</b><br>types the language brings: Date to Secret        | <b>32</b><br>icons, drawn inline, on screen and paper   | <b>1</b><br>binary: compiler, formatter, tests, server               |

# This deck is a .wf file.

```
examples/decks/webfLuent/src/Deck.wf

component Opener(_ title: String, num: String, line: String) {
  Slide(class: "section") {
    Text(num, class: "section__num")
    Text(title, class: "section__title")
    Text(line, class: "section__line")
  }
}

page Deck(path: "/", title: "WebFluent") {
  Presentation {
    Opener("The language", num: "01",
          line: "One lexeme, one meaning.")
    for t in TOPICS { Topic(t) }
  }
}
```

Slides are components: an opener, a heading, a console frame — written once, placed sixteen times. A for makes a slide from each item of a list.

The fonts are files beside it; the colours are the docs site's tokens; wf build writes the PDF.



THE SOURCE OF EVERY DECK

GET STARTED

# Write the page.

## Let the compiler do the rest.

```
$ curl -sSL .../install.sh | bash
```

```
$ wf init my-site --template static
```

```
$ wf serve
```



[monzeromer-lab.github.io/WebFluent](https://monzeromer-lab.github.io/WebFluent)